MAKEGOOD POWER CHAIR

Price Matrix and Reference Build Sheets

Seven costed configurations generated by the configurator, plus the unit price book

R.O.A.R. / TSSFAA with **MakeGood INC**
Generated 2026-10-15 from repository revision **unversioned** · Source: docs/price-matrix.md, docs/examples/A-donated-minimum.md, docs/examples/B-default.md, docs/examples/C-hoverboard-hotswap.md, docs/examples/D-odrive.md, docs/examples/E-wheelchair-brushed.md, docs/examples/F-wheelchair-standard.md, docs/examples/G-full-assist.md, docs/price-book.md
Private pre-release draft. Not yet safe to build from. Prices are September 2026 US retail.

Parts-cost ranges for seven representative configurations, **computed by the configurator's rules engine** (`configurator/index.html` , rules 0.4) — not hand-tallied. Regenerate with `docs/tools/run-scenarios.js` after any change to the price book or rules.

Prices: US retail, September 2026, USD. Ranges exclude shipping, tax, filament, printed parts, and items marked TBD. Basis per line: `listed` = seen in a US retail listing; `est.` = estimate to verify; `donated` = $0. Full build sheets with line items and verification checklists are in `docs/examples/` .

Summary

#	Configuration	Parts low	Parts high	Lines priced	Blocking	Cautions	Build sheet
A	Donated hoverboard, minimum	$511	$1,262	25/25	0	2	A-donated-minimum.md
B	Configurator defaults	$651	$1,502	23/23	0	1	B-default.md
C	New hubs, hot-swap, solenoid brake	$949	$2,228	29/29	0	1	C-hoverboard-hotswap.md
D	New hubs on 2× ODrive S1	$1,081	$2,260	29/30	0	1	D-odrive.md
E	Wheelchair gearmotors + RoboClaw	$1,055	$2,231	21/21	0	1	E-wheelchair-brushed.md
F	Wheelchair motors + standard controller	$850	$2,106	18/18	0	3	F-wheelchair-standard.md
G	Full assist + Sip-N-Puff + buddy seat	$1,533	$3,397	41/42	0	2	G-full-assist.md

What each configuration is

A. Donated hoverboard, minimum — Cheapest viable 36 V build: donated hoverboard, VESC, 2×2 Ah DeWalt, manual wheel locks, no assist, parent e-stop remote.

B. Configurator defaults — What the tool proposes untouched: new hubs, VESC, 2×2 Ah, lift-lever dual mode, BLE remote, key+app settings.

C. New hubs, hot-swap, solenoid brake — Recommended 36 V build for daily use: new 8 in hubs, VESC, 4×5 Ah hot-swap, solenoid pin lock + manual locks, BLE remote.

D. New hubs on 2× ODrive S1 — Same as C but ODrive S1 instead of VESC (brake resistor included, listed price).

E. Wheelchair gearmotors + RoboClaw — 24 V brushed wheelchair gearmotors with integrated EM brake, RoboClaw 2×30A, 24 V 10 Ah pack.

F. Wheelchair motors + standard controller — Most conventional: 24 V gearmotors with a standard wheelchair controller/joystick, 24 V 6 Ah pack, tether kill cord, on-chair settings only — no supervisor-driven remote.

G. Full assist + Sip-N-Puff + buddy seat — Everything on: C plus VENTUNO Q assist (collision, drop-off, tip, power), Sip-N-Puff + joystick input, buddy seat, ventilator on own batteries.

Key choices side by side

Choice	A	B	C	D	E	F	G
Drive	hub36	hub36	hub36	hub36	wcbrushed	wcbrushed	hub36
Motor source	donated	new	new	new	new	new	new
Wheel	8	8	8	8	8	8	8
Controller	vesc	vesc	vesc	odrive	roboclaw	wcctrl	vesc
Battery	2ah	2ah	hot	hot	b24_10	b24_6	hot
Parking brake	manual	manual	manual, pin	manual, pin	integ, manual	integ	manual, pin
Dual mode	lift	lift	lift	lift	lift	lift	lift
Assist computer	none	none	none	none	none	none	ventuno
Assist features	tip, power	tip, power	tip, power	tip, power	tip, power	tip, power	collide, drop, tip, power
Driver input	joy	joy	joy	joy	joy	joy	joy, snp
Supervision	estop, takeover	estop, takeover	estop, takeover	estop, takeover	estop, takeover	tether	estop, takeover
Attendant link	ble	ble	ble	ble	ble	none	ble
Passenger	none	none	none	none	none	none	seat
Ventilator	none	none	none	none	none	none	indep

Values are the configurator's option keys (see `configurator-guide.md` for the key → label table).

Reading the numbers

- **Low** sums every line's low price; **high** sums every high. A real build lands somewhere between, and closer to low if parts are bought carefully and the builder already owns a charger and tools.
- The spread is mostly `est.` uncertainty. As lines move to `listed` the range tightens.

- Configuration A assumes a donated hoverboard that passes the pre-teardown test. A failed donation costs time, not money.
- Wheelchair-motor builds (E, F) look price-competitive with 36 V builds but add 6–14 kg of drive hardware to a toddler chair — see the weight cautions in their build sheets.
- G is the ceiling: vision assist (VENTUNO Q, $299 listed) and a buddy seat sub-frame are the big adders. Sip-N-Puff and head-array inputs are priced TBD (R.O.A.R. modules).

Cost levers, largest first

1. Motor source — donated ($0) vs new hubs ($140–260/pair) vs wheelchair motors ($320–900/pair)
2. Battery capacity — 2×2 Ah ($58–120) vs 4×5 Ah hot-swap ($236–516) vs 24 V 10 Ah ($270–700)
3. Assist compute — none vs UNO Q ($45–70 est.) vs VENTUNO Q ($299 listed)
4. Controller — VESC ($180–260 est.) vs 2× ODrive S1 ($322 listed incl. heat spreaders) vs RoboClaw ($135 listed)
5. Brake strategy — manual locks only ($30–80) vs solenoid pin ($65–170) vs EM brake ($110–320)

A. Donated hoverboard, minimum

Cheapest viable 36 V build: donated hoverboard, VESC, 2×2 Ah DeWalt, manual wheel locks, no assist, parent e-stop remote.

Configuration

- Driver mass (max): Up to 25 kg
- Friend riding along: No passenger
- Manual / power mechanism: Lever-deployed drive wheels
- Drive motors: Hoverboard-class hub
- Motor source: Donated working hoverboard
- Drive wheel size: 8–8.5 in
- Motor controller: VESC dual
- Battery: 2× DeWalt 2 Ah
- Regenerative braking: Off
- Parking brake: Manual wheel locks
- Assistive driving computer: None
- Assist features: Tip / slope monitoring, Power monitoring and range
- Assist default level: Warn only
- Ventilator: None
- Driver input: Joystick
- Supervision: E-stop remote, Attendant takeover
- Attendant link: Bluetooth LE
- Parent settings interface: On-chair key + display

Warnings

- [caution] Donated hoverboard battery: not reused in the chair by default. Its age and cell quality are unknown. If considered, require a UL 2272 label, capacity test, and BMS check; otherwise send it to battery recycling (DRV-2d).
- [caution] VESC clones vary widely. Specify a vetted source and bench-test each unit.
- [info] Counterfeit DeWalt-style packs are common at low prices. Buy genuine packs from authorized retailers.
- [info] Manual locks only: not automatic. The chair holds only when a caregiver applies them.

Parts

Category	Item	Qty	Est. cost (USD)	Price basis	Note
Control	RP2350 supervisor board	1	7–15	est	radio-equipped, e.g. Pico 2 W class
Power	Main fuse or DC breaker, bus-rated	1	10–25	est	
Power	Main power switch, child-inaccessible	1	10–30	est	SAF-3
Power	Battery state display	1	5–20	est	BAT-9
Drive	Donated working hoverboard	1	0–0	donated	source of a matched motor pair; controller still replaced
Drive	Hub motors from donated hoverboard, 8 in, 36 V	2	0–0	donated	wheel size set by the donated unit
Tools	Teardown tools (driver bits, wire cutters, heat shrink)	1	0–20	est	skip if owned
Drive	Metal axle clamp brackets	2	20–60	est	MEC-2
Drive	Motor temperature sensor	2	2–6	est	SAF-9
Control	Dual VESC controller, vetted unit, hall FOC	1	180–260	est	verify rating at 36 V
Control	CAN transceiver, 3.3 V	1	2–8	est	
Power	DeWalt 20V MAX 2 Ah (DCB203, 36 Wh)	2	58–120	listed	genuine only; ≤100 Wh each
Power	DeWalt 20V battery adapter with leads	2	20–50	est	
Power	Series harness, 2 packs to 36 V	1	5–15	est	
Power	DeWalt 20V charger	1	30–70	est	skip if owned
Power	Per-pack voltage sense divider	1	1–3	est	midpoint tap, BAT-4
Mechanical	Lift lever linkage with over-center lock	1	30–100	est	DM-3
Safety	Deploy/lock interlock switch	2	6–20	est	DM-2
Brake	Wheel lock, lever-on-tire	4	60–160	est	printed wheels and drive wheels
Assist	IMU (pitch/roll/rate)	1	5–15	est	
Assist	Bus current/voltage monitor	1	5–15	est	

Category	Item	Qty	Est. cost (USD)	Price basis	Note
Input	Proportional joystick, adjustable mount	1	20–150	est	INP-4
Control	Parent remote (BLE) with e-stop button	1	20–60	est	
Control	Keyswitch for settings lock	1	5–15	est	CFG-3
Control	Small status display	1	10–25	est	CFG-8

Parts total: $511–$1,262 (25 of 25 lines priced; excludes shipping, tax, filament, TBD items). Prices as of Sept 2026; "listed" = seen in a US retail listing, "est." = estimate to verify.

Verification

- [ ] Before teardown: power up the donated hoverboard and confirm both wheels drive smoothly in both directions, with no grinding or hall faults. Record make, model, and any UL 2272 label.
- [ ] After teardown: incoming inspection per DRV-2a/DRV-2d (bearings, all 6 hall states, phase resistance, insulation, tire, axle).
- [ ] Load the versioned controller config file; verify speed cap, current limits, and command timeout (ELE-2a, SAF-5).
- [ ] Stability in manual, powered, and transition positions (DM-8). Lever force with max occupant (DM-5).
- [ ] Brake hold on 1:12 slope, power off, max mass, in every drive mode (SAF-1, DM-7).
- [ ] Run the tip/slope test protocol before enabling above Warn (ASD-12).
- [ ] Run the power test protocol before enabling above Warn (ASD-12).
- [ ] Tip test: IMU beyond recovery angle disables drive and alerts caregiver (ASD-13, proposed).
- [ ] Unplug or fault each input device while driving: chair stops (INP-3).
- [ ] Link-loss test: turn the remote off while driving; chair stops within the set time (ATT-3).

B. Configurator defaults

What the tool proposes untouched: new hubs, VESC, 2×2 Ah, lift-lever dual mode, BLE remote, key+app settings.

Configuration

- Driver mass (max): Up to 25 kg
- Friend riding along: No passenger
- Manual / power mechanism: Lever-deployed drive wheels
- Drive motors: Hoverboard-class hub
- Motor source: New
- Drive wheel size: 8–8.5 in
- Motor controller: VESC dual
- Battery: 2× DeWalt 2 Ah
- Regenerative braking: Off
- Parking brake: Manual wheel locks
- Assistive driving computer: None
- Assist features: Tip / slope monitoring, Power monitoring and range
- Assist default level: Warn only
- Ventilator: None
- Driver input: Joystick
- Supervision: E-stop remote, Attendant takeover
- Attendant link: Bluetooth LE
- Parent settings interface: Both

Warnings

- [caution] VESC clones vary widely. Specify a vetted source and bench-test each unit.
- [info] Counterfeit DeWalt-style packs are common at low prices. Buy genuine packs from authorized retailers.
- [info] Manual locks only: not automatic. The chair holds only when a caregiver applies them.

Parts

Category	Item	Qty	Est. cost (USD)	Price basis	Note
Control	RP2350 supervisor board	1	7–15	est	radio-equipped, e.g. Pico 2 W class
Power	Main fuse or DC breaker, bus-rated	1	10–25	est	
Power	Main power switch, child-inaccessible	1	10–30	est	SAF-3
Power	Battery state display	1	5–20	est	BAT-9
Drive	Hoverboard-class hub motor, 8 in, 36 V, hall-sensored	2	140–260	listed	new
Drive	Metal axle clamp brackets	2	20–60	est	MEC-2
Drive	Motor temperature sensor	2	2–6	est	SAF-9
Control	Dual VESC controller, vetted unit, hall FOC	1	180–260	est	verify rating at 36 V
Control	CAN transceiver, 3.3 V	1	2–8	est	
Power	DeWalt 20V MAX 2 Ah (DCB203, 36 Wh)	2	58–120	listed	genuine only; ≤100 Wh each
Power	DeWalt 20V battery adapter with leads	2	20–50	est	
Power	Series harness, 2 packs to 36 V	1	5–15	est	
Power	DeWalt 20V charger	1	30–70	est	skip if owned
Power	Per-pack voltage sense divider	1	1–3	est	midpoint tap, BAT-4
Mechanical	Lift lever linkage with over-center lock	1	30–100	est	DM-3
Safety	Deploy/lock interlock switch	2	6–20	est	DM-2
Brake	Wheel lock, lever-on-tire	4	60–160	est	printed wheels and drive wheels
Assist	IMU (pitch/roll/rate)	1	5–15	est	
Assist	Bus current/voltage monitor	1	5–15	est	
Input	Proportional joystick, adjustable mount	1	20–150	est	INP-4
Control	Parent remote (BLE) with e-stop button	1	20–60	est	
Control	Keyswitch for settings lock	1	5–15	est	CFG-3
Control	Small status display	1	10–25	est	CFG-8

Parts total: $651–$1,502 (23 of 23 lines priced; excludes shipping, tax, filament, TBD items). Prices as of Sept 2026; "listed" = seen in a US retail listing, "est." = estimate to verify.

Verification

- [ ] Load the versioned controller config file; verify speed cap, current limits, and command timeout (ELE-2a, SAF-5).
- [ ] Stability in manual, powered, and transition positions (DM-8). Lever force with max occupant (DM-5).
- [ ] Brake hold on 1:12 slope, power off, max mass, in every drive mode (SAF-1, DM-7).
- [ ] Run the tip/slope test protocol before enabling above Warn (ASD-12).
- [ ] Run the power test protocol before enabling above Warn (ASD-12).
- [ ] Tip test: IMU beyond recovery angle disables drive and alerts caregiver (ASD-13, proposed).
- [ ] Unplug or fault each input device while driving: chair stops (INP-3).
- [ ] Link-loss test: turn the remote off while driving; chair stops within the set time (ATT-3).

C. New hubs, hot-swap, solenoid brake

Recommended 36 V build for daily use: new 8 in hubs, VESC, 4×5 Ah hot-swap, solenoid pin lock + manual locks, BLE remote.

Configuration

- Driver mass (max): Up to 25 kg
- Friend riding along: No passenger
- Manual / power mechanism: Lever-deployed drive wheels
- Drive motors: Hoverboard-class hub
- Motor source: New
- Drive wheel size: 8–8.5 in
- Motor controller: VESC dual
- Battery: 4× DeWalt 5 Ah hot-swap
- Regenerative braking: Off
- Parking brake: Manual wheel locks, Solenoid pin lock
- Assistive driving computer: None
- Assist features: Tip / slope monitoring, Power monitoring and range
- Assist default level: Warn only
- Ventilator: None
- Driver input: Joystick
- Supervision: E-stop remote, Attendant takeover
- Attendant link: Bluetooth LE
- Parent settings interface: Both

Warnings

- [caution] VESC clones vary widely. Specify a vetted source and bench-test each unit.
- [info] Counterfeit DeWalt-style packs are common at low prices. Buy genuine packs from authorized retailers.
- [info] Solenoid pin lock engages at zero speed only. The supervisor must enforce this (SAF-2).

Parts

Category	Item	Qty	Est. cost (USD)	Price basis	Note
Control	RP2350 supervisor board	1	7–15	est	radio-equipped, e.g. Pico 2 W class
Power	Main fuse or DC breaker, bus-rated	1	10–25	est	
Power	Main power switch, child-inaccessible	1	10–30	est	SAF-3
Power	Battery state display	1	5–20	est	BAT-9
Drive	Hoverboard-class hub motor, 8 in, 36 V, hall-sensored	2	140–260	listed	new
Drive	Metal axle clamp brackets	2	20–60	est	MEC-2
Drive	Motor temperature sensor	2	2–6	est	SAF-9
Control	Dual VESC controller, vetted unit, hall FOC	1	180–260	est	verify rating at 36 V
Control	CAN transceiver, 3.3 V	1	2–8	est	
Power	DeWalt 20V MAX 5 Ah (DCB205, 90 Wh)	4	236–516	listed	genuine only; ≤100 Wh each
Power	DeWalt 20V battery adapter with leads	4	40–100	est	
Power	Series harness, 2 packs to 36 V	2	10–30	est	
Power	DeWalt 20V charger	1	30–70	est	skip if owned
Power	Per-pack voltage sense divider	1	1–3	est	midpoint tap, BAT-4
Power	Ideal-diode controller + MOSFETs (LM74700-class)	2	20–50	est	BAT-6
Power	Bus brake / dump resistor with switch	1	10–30	est	SAF-10
Mechanical	Lift lever linkage with over-center lock	1	30–100	est	DM-3
Safety	Deploy/lock interlock switch	2	6–20	est	DM-2
Brake	Wheel lock, lever-on-tire	4	60–160	est	printed wheels and drive wheels
Brake	Push-pull solenoid, spring-engaged	2	30–80	est	36 V
Brake	Indexed lock disc for wheel	2	20–60	est	
Brake	Hit-and-hold solenoid/actuator driver	1	5–15	est	SAF-2a
Brake	Manual brake release	1	10–30	est	SAF-2c
Assist	IMU (pitch/roll/rate)	1	5–15	est	

Category	Item	Qty	Est. cost (USD)	Price basis	Note
Assist	Bus current/voltage monitor	1	5–15	est	
Input	Proportional joystick, adjustable mount	1	20–150	est	INP-4
Control	Parent remote (BLE) with e-stop button	1	20–60	est	
Control	Keyswitch for settings lock	1	5–15	est	CFG-3
Control	Small status display	1	10–25	est	CFG-8

Parts total: $949–$2,228 (29 of 29 lines priced; excludes shipping, tax, filament, TBD items). Prices as of Sept 2026; "listed" = seen in a US retail listing, "est." = estimate to verify.

Verification

- [ ] Load the versioned controller config file; verify speed cap, current limits, and command timeout (ELE-2a, SAF-5).
- [ ] Hot-swap test: remove one pair while driving; bus must stay up.
- [ ] Stability in manual, powered, and transition positions (DM-8). Lever force with max occupant (DM-5).
- [ ] Add-on brakes engage on power loss and on e-stop; manual release returns safely (SAF-2b, SAF-2c).
- [ ] Brake hold on 1:12 slope, power off, max mass, in every drive mode (SAF-1, DM-7).
- [ ] Run the tip/slope test protocol before enabling above Warn (ASD-12).
- [ ] Run the power test protocol before enabling above Warn (ASD-12).
- [ ] Tip test: IMU beyond recovery angle disables drive and alerts caregiver (ASD-13, proposed).
- [ ] Unplug or fault each input device while driving: chair stops (INP-3).
- [ ] Link-loss test: turn the remote off while driving; chair stops within the set time (ATT-3).

D. New hubs on 2× ODrive S1

Same as C but ODrive S1 instead of VESC (brake resistor included, listed price).

Configuration

- Driver mass (max): Up to 25 kg
- Friend riding along: No passenger
- Manual / power mechanism: Lever-deployed drive wheels
- Drive motors: Hoverboard-class hub
- Motor source: New
- Drive wheel size: 8–8.5 in
- Motor controller: 2× ODrive S1
- Battery: 4× DeWalt 5 Ah hot-swap
- Regenerative braking: Off
- Parking brake: Manual wheel locks, Solenoid pin lock
- Assistive driving computer: None
- Assist features: Tip / slope monitoring, Power monitoring and range
- Assist default level: Warn only
- Ventilator: None
- Driver input: Joystick
- Supervision: E-stop remote, Attendant takeover
- Attendant link: Bluetooth LE
- Parent settings interface: Both

Warnings

- [caution] ODrive S1 bus limit is 50.5 V vs 42 V full 10S. Keep brake resistors fitted even with regen off.
- [info] Counterfeit DeWalt-style packs are common at low prices. Buy genuine packs from authorized retailers.
- [info] Solenoid pin lock engages at zero speed only. The supervisor must enforce this (SAF-2).

Parts

Category	Item	Qty	Est. cost (USD)	Price basis	Note
Control	RP2350 supervisor board	1	7–15	est	radio-equipped, e.g. Pico 2 W class
Power	Main fuse or DC breaker, bus-rated	1	10–25	est	
Power	Main power switch, child-inaccessible	1	10–30	est	SAF-3
Power	Battery state display	1	5–20	est	BAT-9
Drive	Hoverboard-class hub motor, 8 in, 36 V, hall-sensored	2	140–260	listed	new
Drive	Metal axle clamp brackets	2	20–60	est	MEC-2
Drive	Motor temperature sensor	2	2–6	est	SAF-9
Control	ODrive S1	2	298–298	listed	12–50.5 V, brake resistor included
Control	ODrive S1 heat spreader plate	2	24–24	listed	
Control	CAN transceiver, 3.3 V	1	2–8	est	
Power	DeWalt 20V MAX 5 Ah (DCB205, 90 Wh)	4	236–516	listed	genuine only; ≤100 Wh each
Power	DeWalt 20V battery adapter with leads	4	40–100	est	
Power	Series harness, 2 packs to 36 V	2	10–30	est	
Power	DeWalt 20V charger	1	30–70	est	skip if owned
Power	Per-pack voltage sense divider	1	1–3	est	midpoint tap, BAT-4
Power	Ideal-diode controller + MOSFETs (LM74700-class)	2	20–50	est	BAT-6
Power	Bus brake / dump resistor with switch	1	TBD		included with ODrive S1
Mechanical	Lift lever linkage with over-center lock	1	30–100	est	DM-3
Safety	Deploy/lock interlock switch	2	6–20	est	DM-2
Brake	Wheel lock, lever-on-tire	4	60–160	est	printed wheels and drive wheels
Brake	Push-pull solenoid, spring-engaged	2	30–80	est	36 V
Brake	Indexed lock disc for wheel	2	20–60	est	
Brake	Hit-and-hold solenoid/actuator driver	1	5–15	est	SAF-2a
Brake	Manual brake release	1	10–30	est	SAF-2c

Category	Item	Qty	Est. cost (USD)	Price basis	Note
Assist	IMU (pitch/roll/rate)	1	5–15	est	
Assist	Bus current/voltage monitor	1	5–15	est	
Input	Proportional joystick, adjustable mount	1	20–150	est	INP-4
Control	Parent remote (BLE) with e-stop button	1	20–60	est	
Control	Keyswitch for settings lock	1	5–15	est	CFG-3
Control	Small status display	1	10–25	est	CFG-8

Parts total: $1,081–$2,260 (29 of 30 lines priced; excludes shipping, tax, filament, TBD items). Prices as of Sept 2026; "listed" = seen in a US retail listing, "est." = estimate to verify.

Verification

- [ ] Load the versioned controller config file; verify speed cap, current limits, and command timeout (ELE-2a, SAF-5).
- [ ] Hot-swap test: remove one pair while driving; bus must stay up.
- [ ] Stability in manual, powered, and transition positions (DM-8). Lever force with max occupant (DM-5).
- [ ] Add-on brakes engage on power loss and on e-stop; manual release returns safely (SAF-2b, SAF-2c).
- [ ] Brake hold on 1:12 slope, power off, max mass, in every drive mode (SAF-1, DM-7).
- [ ] Run the tip/slope test protocol before enabling above Warn (ASD-12).
- [ ] Run the power test protocol before enabling above Warn (ASD-12).
- [ ] Tip test: IMU beyond recovery angle disables drive and alerts caregiver (ASD-13, proposed).
- [ ] Unplug or fault each input device while driving: chair stops (INP-3).
- [ ] Link-loss test: turn the remote off while driving; chair stops within the set time (ATT-3).

E. Wheelchair gearmotors + RoboClaw

24 V brushed wheelchair gearmotors with integrated EM brake, RoboClaw 2×30A, 24 V 10 Ah pack.

Configuration

- Driver mass (max): Up to 25 kg
- Friend riding along: No passenger
- Manual / power mechanism: Lever-deployed drive wheels
- Drive motors: Wheelchair gearmotor
- Motor source: New
- Drive wheel size: 8–8.5 in
- Motor controller: RoboClaw 2×30A
- Battery: 24 V Li-ion 10 Ah
- Regenerative braking: Off
- Parking brake: Motor's built-in EM brake, Manual wheel locks
- Assistive driving computer: None
- Assist features: Tip / slope monitoring, Power monitoring and range
- Assist default level: Warn only
- Ventilator: None
- Driver input: Joystick
- Supervision: E-stop remote, Attendant takeover
- Attendant link: Bluetooth LE
- Parent settings interface: Both

Warnings

- [caution] Wheelchair gearmotors weigh about 3 kg each or more. Recheck CG, tip stability, and lift-lever force for a toddler chair.
- [info] Some vendors offer an encoder only in place of the brake. Confirm a model with both, or fit an external encoder.
- [info] 240 Wh is over 160 Wh. Airline rules allow it as an installed mobility-aid battery, but limit spares (REG-6).
- [info] Branded chair packs listed at the high end of the range; generic packs cost less but check BMS quality and airline paperwork.

Parts

Category	Item	Qty	Est. cost (USD)	Price basis	Note
Control	RP2350 supervisor board	1	7–15	est	radio-equipped, e.g. Pico 2 W class
Power	Main fuse or DC breaker, bus-rated	1	10–25	est	
Power	Main power switch, child-inaccessible	1	10–30	est	SAF-3
Power	Battery state display	1	5–20	est	BAT-9
Drive	Wheelchair gearmotor, 24 V brushed, integrated EM brake + freewheel	2	320–400	listed	e.g. 250 W class, 17 mm keyed shaft
Drive	Drive wheel with keyed hub, 8 in	2	80–180	est	
Drive	Motor mounting brackets	2	20–60	est	
Control	Wheel/shaft encoder for closed-loop speed	2	40–120	est	ELE-2
Drive	Motor temperature sensor	2	2–6	est	SAF-9
Control	RoboClaw 2×30A	1	135–135	listed	6–34 V, encoder speed control, regen
Power	24 V Li-ion pack, 10 Ah (240 Wh), removable, UN38.3 docs	1	270–700	listed	
Power	24 V Li-ion charger (29.4 V)	1	25–60	est	
Mechanical	Lift lever linkage with over-center lock	1	30–100	est	DM-3
Safety	Deploy/lock interlock switch	2	6–20	est	DM-2
Brake	Wheel lock, lever-on-tire	2	30–80	est	
Assist	IMU (pitch/roll/rate)	1	5–15	est	
Assist	Bus current/voltage monitor	1	5–15	est	
Input	Proportional joystick, adjustable mount	1	20–150	est	INP-4
Control	Parent remote (BLE) with e-stop button	1	20–60	est	
Control	Keyswitch for settings lock	1	5–15	est	CFG-3
Control	Small status display	1	10–25	est	CFG-8

Parts total: $1,055–$2,231 (21 of 21 lines priced; excludes shipping, tax, filament, TBD items). Prices as of Sept 2026; "listed" = seen in a US retail listing, "est." = estimate to verify.

Verification

- [ ] Load the versioned controller config file; verify speed cap, current limits, and command timeout (ELE-2a, SAF-5).
- [ ] Stability in manual, powered, and transition positions (DM-8). Lever force with max occupant (DM-5).
- [ ] Manual mode: motor brakes released or drive wheels lifted, with no drag on the printed wheels (DM-6).
- [ ] Motor EM brakes engage on power loss and e-stop; freewheel/manual release returns safely (SAF-2b, SAF-2c).
- [ ] Brake hold on 1:12 slope, power off, max mass, in every drive mode (SAF-1, DM-7).
- [ ] Run the tip/slope test protocol before enabling above Warn (ASD-12).
- [ ] Run the power test protocol before enabling above Warn (ASD-12).
- [ ] Tip test: IMU beyond recovery angle disables drive and alerts caregiver (ASD-13, proposed).
- [ ] Unplug or fault each input device while driving: chair stops (INP-3).
- [ ] Link-loss test: turn the remote off while driving; chair stops within the set time (ATT-3).

F. Wheelchair motors + standard controller

Most conventional: 24 V gearmotors with a standard wheelchair controller/joystick, 24 V 6 Ah pack, tether kill cord, on-chair settings only — no supervisor-driven remote.

Configuration

- Driver mass (max): Up to 25 kg
- Friend riding along: No passenger
- Manual / power mechanism: Lever-deployed drive wheels
- Drive motors: Wheelchair gearmotor
- Motor source: New
- Drive wheel size: 8–8.5 in
- Motor controller: Standard wheelchair controller
- Battery: 24 V Li-ion 6 Ah
- Regenerative braking: Off
- Parking brake: Motor's built-in EM brake
- Assistive driving computer: None
- Assist features: Tip / slope monitoring, Power monitoring and range
- Assist default level: Warn only
- Ventilator: None
- Driver input: Joystick
- Supervision: Tether kill cord
- Attendant link: None
- Parent settings interface: On-chair key + display

Warnings

- [caution] Wheelchair gearmotors weigh about 3 kg each or more. Recheck CG, tip stability, and lift-lever force for a toddler chair.
- [caution] Standard wheelchair controllers are usually proprietary and often need a dealer programmer. Alternate inputs, parent remotes, and profiles (4.6) only work if the controller exposes an input the supervisor can drive.
- [caution] Motor brakes hold only in power mode. Add wheel locks for the printed wheels in manual mode (DM-7).
- [info] Branded chair packs listed at the high end of the range; generic packs cost less but check BMS quality and airline paperwork.

Parts

Category	Item	Qty	Est. cost (USD)	Price basis	Note
Control	RP2350 supervisor board	1	7–15	est	
Power	Main fuse or DC breaker, bus-rated	1	10–25	est	
Power	Main power switch, child-inaccessible	1	10–30	est	SAF-3
Power	Battery state display	1	5–20	est	BAT-9
Drive	Wheelchair gearmotor, 24 V brushed, integrated EM brake + freewheel	2	320–400	listed	e.g. 250 W class, 17 mm keyed shaft
Drive	Drive wheel with keyed hub, 8 in	2	80–180	est	
Drive	Motor mounting brackets	2	20–60	est	
Drive	Motor temperature sensor	2	2–6	est	SAF-9
Control	Standard wheelchair controller with joystick and brake outputs	1	150–600	est	verify model
Power	24 V Li-ion pack, 6 Ah (144 Wh), removable, UN38.3 docs	1	150–495	listed	
Power	24 V Li-ion charger (29.4 V)	1	25–60	est	
Mechanical	Lift lever linkage with over-center lock	1	30–100	est	DM-3
Safety	Deploy/lock interlock switch	2	6–20	est	DM-2
Assist	IMU (pitch/roll/rate)	1	5–15	est	
Assist	Bus current/voltage monitor	1	5–15	est	
Safety	Magnetic tether kill switch with lanyard	1	10–25	est	
Control	Keyswitch for settings lock	1	5–15	est	CFG-3
Control	Small status display	1	10–25	est	CFG-8

Parts total: $850–$2,106 (18 of 18 lines priced; excludes shipping, tax, filament, TBD items). Prices as of Sept 2026; "listed" = seen in a US retail listing, "est." = estimate to verify.

Verification

- [ ] Stability in manual, powered, and transition positions (DM-8). Lever force with max occupant (DM-5).
- [ ] Manual mode: motor brakes released or drive wheels lifted, with no drag on the printed wheels (DM-6).

- [ ] Motor EM brakes engage on power loss and e-stop; freewheel/manual release returns safely (SAF-2b, SAF-2c).
- [ ] Brake hold on 1:12 slope, power off, max mass, in every drive mode (SAF-1, DM-7).
- [ ] Run the tip/slope test protocol before enabling above Warn (ASD-12).
- [ ] Run the power test protocol before enabling above Warn (ASD-12).
- [ ] Tip test: IMU beyond recovery angle disables drive and alerts caregiver (ASD-13, proposed).
- [ ] Unplug or fault each input device while driving: chair stops (INP-3).
- [ ] Pull tether while driving: chair stops.

G. Full assist + Sip-N-Puff + buddy seat

Everything on: C plus VENTUNO Q assist (collision, drop-off, tip, power), Sip-N-Puff + joystick input, buddy seat, ventilator on own batteries.

Configuration

- Driver mass (max): Up to 25 kg
- Friend riding along: Buddy seat
- Manual / power mechanism: Lever-deployed drive wheels
- Drive motors: Hoverboard-class hub
- Motor source: New
- Drive wheel size: 8–8.5 in
- Motor controller: VESC dual
- Battery: 4× DeWalt 5 Ah hot-swap
- Regenerative braking: Off
- Parking brake: Manual wheel locks, Solenoid pin lock
- Assistive driving computer: Arduino VENTUNO Q
- Assist features: Collision avoidance, Drop-off / stair / curb edge, Tip / slope monitoring, Power monitoring and range
- Assist default level: Warn + slow
- Ventilator: On its own batteries
- Driver input: Joystick, Sip-N-Puff
- Supervision: E-stop remote, Attendant takeover
- Attendant link: Bluetooth LE
- Parent settings interface: Both

Warnings

- [caution] VESC clones vary widely. Specify a vetted source and bench-test each unit.
- [caution] Passenger with ventilator mounted is a parent setting, default off. Verify stability for this combination first (PAS-5).
- [info] Counterfeit DeWalt-style packs are common at low prices. Buy genuine packs from authorized retailers.
- [info] Solenoid pin lock engages at zero speed only. The supervisor must enforce this (SAF-2).
- [info] Assist is not part of the Safety Floor. The caregiver guide must state it cannot detect all hazards and may false-alarm (ASD-11).

Parts

Category	Item	Qty	Est. cost (USD)	Price basis	Note
Control	RP2350 supervisor board	1	7–15	est	radio-equipped, e.g. Pico 2 W class
Power	Main fuse or DC breaker, bus-rated	1	10–25	est	
Power	Main power switch, child-inaccessible	1	10–30	est	SAF-3
Power	Battery state display	1	5–20	est	BAT-9
Drive	Hoverboard-class hub motor, 8 in, 36 V, hall-sensored	2	140–260	listed	new
Drive	Metal axle clamp brackets	2	20–60	est	MEC-2
Drive	Motor temperature sensor	2	2–6	est	SAF-9
Control	Dual VESC controller, vetted unit, hall FOC	1	180–260	est	verify rating at 36 V
Control	CAN transceiver, 3.3 V	1	2–8	est	
Power	DeWalt 20V MAX 5 Ah (DCB205, 90 Wh)	4	236–516	listed	genuine only; ≤100 Wh each
Power	DeWalt 20V battery adapter with leads	4	40–100	est	
Power	Series harness, 2 packs to 36 V	2	10–30	est	
Power	DeWalt 20V charger	1	30–70	est	skip if owned
Power	Per-pack voltage sense divider	1	1–3	est	midpoint tap, BAT-4
Power	Ideal-diode controller + MOSFETs (LM74700-class)	2	20–50	est	BAT-6
Power	Bus brake / dump resistor with switch	1	10–30	est	SAF-10
Safety	Passenger seat sensor	1	5–20	est	PAS-2
Safety	Passenger restraint with latch switch	1	20–60	est	PAS-3
Mechanical	Metal sub-frame for buddy seat	1	50–150	est	MEC-3
Mechanical	Buddy seat and guards	1	50–200	est	PAS-4, PAS-7 width
Mechanical	Lift lever linkage with over-center lock	1	30–100	est	DM-3
Safety	Deploy/lock interlock switch	2	6–20	est	DM-2
Brake	Wheel lock, lever-on-tire	4	60–160	est	printed wheels and drive wheels

Category	Item	Qty	Est. cost (USD)	Price basis	Note
Brake	Push-pull solenoid, spring-engaged	2	30–80	est	36 V
Brake	Indexed lock disc for wheel	2	20–60	est	
Brake	Hit-and-hold solenoid/actuator driver	1	5–15	est	SAF-2a
Brake	Manual brake release	1	10–30	est	SAF-2c
Assist	Arduino VENTUNO Q	1	299–299	listed	Dragonwing IQ8 + STM32H5, 3× MIPI-CSI
Assist	Regulated DC-DC for assist computer from 36 V bus	1	15–40	est	ASD-14
Assist	Assist override switch (matched to driver input)	1	25–70	est	ASD-3
Assist	Warning annunciator (tones, LEDs, haptic)	1	5–20	est	ASD-8
Assist	Camera module (MIPI-CSI)	1	25–60	est	on-device processing only (ASD-9)
Assist	ToF/ultrasonic ranging sensor (front, rear, sides)	4	40–100	est	
Assist	Downward ToF cliff sensor	2	20–50	est	front corners
Assist	IMU (pitch/roll/rate)	1	5–15	est	
Assist	Bus current/voltage monitor	1	5–15	est	
Mechanical	Ventilator mount/tray with tubing guard	1	30–100	est	VNT-8
Input	Proportional joystick, adjustable mount	1	20–150	est	INP-4
Input	R.O.A.R. Sip-N-Puff module	1	TBD		
Control	Parent remote (BLE) with e-stop button	1	20–60	est	
Control	Keyswitch for settings lock	1	5–15	est	CFG-3
Control	Small status display	1	10–25	est	CFG-8

Parts total: $1,533–$3,397 (41 of 42 lines priced; excludes shipping, tax, filament, TBD items). Prices as of Sept 2026; "listed" = seen in a US retail listing, "est." = estimate to verify.

Verification

- [ ] Load the versioned controller config file; verify speed cap, current limits, and command timeout (ELE-2a, SAF-5).
- [ ] Hot-swap test: remove one pair while driving; bus must stay up.
- [ ] Brake hold, ramp climb, stability, and runtime at max combined mass (PAS-1, PAS-6, PAS-8).
- [ ] Stability in manual, powered, and transition positions (DM-8). Lever force with max occupant (DM-5).
- [ ] Add-on brakes engage on power loss and on e-stop; manual release returns safely (SAF-2b, SAF-2c).
- [ ] Brake hold on 1:12 slope, power off, max mass, in every drive mode (SAF-1, DM-7).
- [ ] Pull assist computer power while driving: limits ramp off, no abrupt stop, caregiver alerted (ASD-5).
- [ ] Driver override works with every configured input method (ASD-3).
- [ ] Run the collision test protocol before enabling above Warn (ASD-12).
- [ ] Run the drop-off test protocol before enabling above Warn (ASD-12).
- [ ] Run the tip/slope test protocol before enabling above Warn (ASD-12).
- [ ] Run the power test protocol before enabling above Warn (ASD-12).
- [ ] Tip test: IMU beyond recovery angle disables drive and alerts caregiver (ASD-13, proposed).
- [ ] Stability with ventilator mounted (VNT-9). EMC check: vent operation near motor controller (VNT-7).
- [ ] Unplug or fault each input device while driving: chair stops (INP-3).
- [ ] Link-loss test: turn the remote off while driving; chair stops within the set time (ATT-3).

Price Book

Unit prices (USD, per item) used by the configurator. This is a rendered view of the `P` table in `configurator/index.html`; **edit the configurator, not this file**, then regenerate with `docs/tools/pricebook.js`.

Basis: `listed` = seen in a US retail listing, September 2026 · `est` = estimate, verify before relying on it · `donated` = $0 · TBD = priced by the vendor at build time.

Rule of thumb for updating: a line moves from `est` to `listed` only with a retailer URL and date noted in the commit message.

Drive

Key	Item	Low	High	Basis
`hubNew`	Hoverboard-class hub motor, new	$70	$130	listed
`hubSalv`	Hoverboard-class hub motor, salvaged	$15	$40	est
`donated`	Donated working hoverboard	$0	$0	donated
`hubDon`	Hub motors from donated hoverboard	$0	$0	donated
`axleBr`	Metal axle clamp brackets	$10	$30	est
`therm`	Motor temperature sensor	$1	$3	est
`wcBrushed`	Wheelchair gearmotor, 24 V brushed, EM brake + freewheel, new	$160	$200	listed
`wcBrushedSalv`	Wheelchair gearmotor, salvaged	$40	$120	est
`wcBldc`	Wheelchair brushless hub wheel, 24 V, EM brake, new	$250	$450	est
`wcBldcSalv`	Wheelchair brushless hub wheel, salvaged	$60	$150	est
`wcWheel`	Drive wheel with keyed hub, … in	$40	$90	est

Control

Key	Item	Low	High	Basis
`rp2350`	RP2350 supervisor board	$7	$15	est
`vesc`	Dual VESC controller, vetted unit, hall FOC	$180	$260	est
`can`	CAN transceiver, 3.3 V	$2	$8	est
`odrive`	ODrive S1	$149	$149	listed
`odHeat`	ODrive S1 heat spreader plate	$12	$12	listed

Key	Item	Low	High	Basis
zs	ZS-X11H hall BLDC driver	$15	$30	est
hbBoard	Hoverboard mainboard (STM32F103)	$20	$40	est
encoder	Wheel/shaft encoder for closed-loop speed	$20	$60	est
roboclaw	RoboClaw 2×30A	$135	$135	listed
sabertooth	Sabertooth 2×32	$125	$125	listed
wcCtrl	Standard wheelchair controller with joystick and brake outputs	$150	$600	est
bleRemote	Parent remote (BLE) with e-stop button	$20	$60	est
rc	2.4 GHz RC transmitter + receiver with failsafe	$40	$120	est
keysw	Keyswitch for settings lock	$5	$15	est
disp	Small status display	$10	$25	est

Power

Key	Item	Low	High	Basis
fuse	Main fuse or DC breaker, bus-rated	$10	$25	est
mainsw	Main power switch, child-inaccessible	$10	$30	est
sense	Per-pack voltage sense divider	$1	$3	est
battdisp	Battery state display	$5	$20	est
dcb203	DeWalt 20V MAX 2 Ah (DCB203)	$29	$60	listed
dcb205	DeWalt 20V MAX 5 Ah (DCB205)	$59	$129	listed
dwAdapt	DeWalt 20V battery adapter with leads	$10	$25	est
dwHarn	Series harness, 2 packs to 36 V	$5	$15	est
dwChg	DeWalt 20V charger	$30	$70	est
idealD	Ideal-diode controller + MOSFETs (LM74700-class)	$10	$25	est
dump	Bus brake / dump resistor with switch	$10	$30	est
b24_6	24 V Li-ion pack, 6 Ah (144 Wh), removable, UN38.3 docs	$150	$495	listed
b24_10	24 V Li-ion pack, 10 Ah (240 Wh), removable, UN38.3 docs	$270	$700	listed
sla12	12 V SLA/AGM 12 Ah, non-spillable	$30	$60	est
chg24	24 V Li-ion charger	$25	$60	est

Key	Item	Low	High	Basis
chgSla	24 V SLA charger	$25	$50	est
dcdc	Isolated DC-DC converter within the vent's labeled DC input range	$60	$200	est
ventCable	Ventilator manufacturer DC cable	TBD	TBD	vendor
ventFuse	Separate vent-feed fuse and switch	$10	$25	est

Brake

Key	Item	Low	High	Basis
wlock	Wheel lock, lever-on-tire	$15	$40	est
sol	Push-pull solenoid, spring-engaged	$15	$40	est
idxDisc	Indexed lock disc for wheel	$10	$30	est
emBrake	Spring-applied, solenoid-released friction brake	$40	$120	est
rotorAd	Rotor/drum adapter	$15	$40	est
caliper	Bicycle disc caliper + rotor	$25	$60	est
springAct	Spring return + release actuator	$20	$60	est
hitHold	Brake coil driver output (hit-and-hold)	$5	$15	est
manRel	Manual brake release	$10	$30	est

Mechanical

Key	Item	Low	High	Basis
subframe	Metal sub-frame for buddy seat	$50	$150	est
buddy	Buddy seat and guards	$50	$200	est
hitch	Trailer hitch, load-rated, into metal bracket	$20	$50	est
trailer	Passenger trailer with seat	$80	$200	est
board	Rear ride-on board with handle	$40	$100	est
lever	Lift lever linkage with over-center lock	$30	$100	est
pod	Power pod frame with quick-release mounts	$50	$150	est
fricArm	Spring-loaded friction drive arm	$20	$60	est
ventMount	Ventilator mount/tray with tubing guard	$30	$100	est

Safety

Key	Item	Low	High	Basis
`seatSens`	Passenger seat sensor	$5	$20	est
`restraint`	Passenger restraint with latch switch	$20	$60	est
`interlock`	Deploy/lock interlock switch	$3	$10	est
`rimGuard`	Push-rim guards	$10	$30	est
`tether`	Magnetic tether kill switch with lanyard	$10	$25	est

Input

Key	Item	Low	High	Basis
`joy`	Proportional joystick, adjustable mount	$20	$150	est
`adSwitch`	Driver enable switch (adapted switch)	$25	$70	est
`jack`	3.5 mm switch jacks	$2	$5	est
`scanDisp`	Scan indicator display or LEDs	$10	$30	est
`snp`	R.O.A.R. Sip-N-Puff module	TBD	TBD	vendor
`head`	Head array	TBD	TBD	vendor

Assist

Key	Item	Low	High	Basis
`ventuno`	Arduino VENTUNO Q	$299	$299	listed
`unoq`	Arduino UNO Q	$45	$70	est
`otherAI`	Sponsor assist computer	$100	$400	est
`cam`	Camera module (MIPI-CSI)	$25	$60	est
`tof`	ToF/ultrasonic ranging sensor (front, rear, sides)	$10	$25	est
`cliff`	Downward ToF cliff sensor	$10	$25	est
`imu`	IMU (pitch/roll/rate)	$5	$15	est
`pwrMon`	Bus current/voltage monitor	$5	$15	est
`annunc`	Warning annunciator (tones, LEDs, haptic)	$5	$20	est

Key	Item	Low	High	Basis
`ovr`	Assist override switch (matched to driver input)	$25	$70	est
`dcdc12`	Regulated DC-DC for assist computer from … V bus	$15	$40	est

Tools

Key	Item	Low	High	Basis
`teardownTools`	Teardown tools (driver bits, wire cutters, heat shrink)	$0	$20	est
`inspect`	Incoming inspection: hall/encoder tester, milliohm meter, insulation check	$30	$80	est
`stlink`	ST-Link programmer	$10	$25	est

Coverage

88 price-book entries: 11 listed, 72 estimated, 2 donated, 3 TBD. Priority for verification: the `est` lines with the largest high values (VESC, wheelchair controller, 24 V packs, EM brakes, buddy-seat sub-frame).